

Rakshya U. Sharma

Curriculum Vitae

San Francisco, CA

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in RakshyaUSharma

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Research Profile

Bioinformatician and quantitative research analyst with experience in computational biology, statistical genetics, epidemiologic data analysis, and reproducible research workflows. My work focuses on integrating large-scale genomic, clinical, phenotypic, and cohort datasets to study disease risk, prevention, and health-related outcomes. I have experience with R, Python, SQL, Bash, Snakemake, Nextflow, Docker, Conda, and high-performance computing environments, with strengths in data harmonization, quality control, statistical modeling, reproducible reporting, and collaborative manuscript-oriented analyses. I am especially interested in applying computational and quantitative methods to population health, prevention science, aging, neurodegenerative disease, health disparities, implementation research, and precision/public health. I am also interested in supporting studies that integrate clinical, behavioral, genetic, EHR, and social-contextual data to identify risk patterns, evaluate interventions, and improve health outcomes in diverse populations.

Research Experience

06/2024–
Present **Bioinformatician**, *Andrews Lab, University of California, San Francisco*, San Francisco, CA,
Department of Psychiatry and Behavioral Sciences.

- Conduct computational analyses for Alzheimer's disease and neurodegenerative disease research using large-scale genomic, clinical, and phenotypic datasets from ADGC and HABS-HD cohorts.
- Designed and implemented the *PRS-CS Auto pipeline* for polygenic risk score generation, quality control, and downstream association analyses across **25k+ samples**, supporting reproducible evaluation of genetic liability to modifiable risk factors in Alzheimer's disease.
- Developed and maintained reproducible Snakemake and Nextflow workflows for high-throughput genomic processing, including genotype QC, score generation, harmonization, and output validation across HPC environments.
- Built and contributed to automated workflows for Mendelian randomization, genetic correlation analysis, and sex-stratified association testing to evaluate genetic and causal relationships between modifiable traits and Alzheimer's disease.
- Integrated and harmonized genomic, clinical, and phenotype metadata from **30+ datasets**, enabling consistent cohort QC, association modeling, and downstream statistical analyses.
- Containerized bioinformatics workflows using Docker and Conda, improving portability, reproducibility, and version control across collaborators and computing environments.
- Collaborated with bioinformaticians, statisticians, clinicians, and neuroscientists to translate biological questions into computational workflows, interpretable results, and manuscript-oriented analyses.
- Performed data validation, discrepancy resolution, and manual quality control on integrated clinical and genomic datasets; supported dataset preparation, reporting, and collaborative research requests.

- 04/2022–06/2024 **Undergraduate Research Assistant**, *Brooks Lab, UCSC Genomics Institute*, Santa Cruz, CA, PI: Angela Brooks, Ph.D..
- Developed the *RBPSig* RNA-seq pipeline using STAR and MESA to generate RNA-binding protein knockdown signatures from ENCODE data and evaluate splicing alterations in TCGA lung adenocarcinoma cohorts.
 - Automated large-scale RNA-seq data ingestion, alignment, splicing quantification, and metadata integration, improving reproducibility across transcriptomic analyses.
 - Applied statistical modeling and multiple testing correction to link splicing signatures with patient-level genomic and clinical data.
 - Validated pipeline performance by reproducing known SUGP1/SF3B1 splicing overlap in LUAD and communicated findings through posters and research reports.
- 05/2022–07/2022 **Broad Summer Research Internship**, *Translational Genomics Group, Broad Institute of MIT and Harvard*, Cambridge, MA, PI: Heidi Rehm, Ph.D..
- Developed a pipeline *broadinstitute/sma-finder* to detect Survival Motor Neuron deletions and mutations from whole-exome sequencing data, supporting improved identification of Spinal Muscular Atrophy (SMA) cases.
 - Analyzed and visualized genomic data using Integrative Genomics Viewer (IGV), improving interpretation of complex variant patterns.
 - Contributed to a peer-reviewed publication through data analysis and pipeline development.
- 06/2021–08/2021 **Community College Internship**, *IDEAL team, Lawrence Berkeley National Lab*, Berkeley, CA, PI: Daniela Ushizima, Ph.D..
- Applied machine learning methods to automate image segmentation of plant root systems, improving accuracy and efficiency of feature extraction.
 - Generated reproducible analysis outputs and presented findings at the Lawrence Berkeley National Lab Poster Symposium 2021.

Education and Trainings

- 06/2024 **Bachelor of Science in Computer Science, Minor in Bioinformatics**, *University of California, Santa Cruz*, Santa Cruz, CA.
Graduated June 2024
- 07/2024–Present **Post-Baccalaureate Research Opportunity to Promote Equity in Learning (PROPEL)**, *University of California, San Francisco*, San Francisco, CA.

Technical Skills

- Programming & Analysis** R, Python, SQL, Bash/Linux scripting, statistical modeling, data wrangling, reproducible reporting
- Bioinformatics** Polygenic risk scores, Mendelian randomization, genetic correlation analysis, genotype/phenotype harmonization, GWAS summary statistics processing, RNA-seq analysis, splicing analysis, variant interpretation
- Workflow Systems** Snakemake, Nextflow, Docker, Conda, Git, Jupyter, RMarkdown; reproducible pipeline development and workflow documentation
- High-Dimensional Data** Experience with large-scale genomic, clinical, and phenotypic datasets; background in RNA-seq and WES analysis; strong interest in applying single-cell RNA-seq, WGS/genotype data, proteomics, and digital pathology to neurodegenerative disease
- HPC & Cloud** SGE, SLURM, UCSF Wynton HPC, AWS, GCP; scalable workflow execution, job submission, and data management in research computing environments

Visualization & Reporting	ggplot2, RMarkdown, IGV, UCSC Genome Browser; publication-oriented plots, statistical summaries, and reproducible analysis reports
Data Management	Clinical/genomic data integration, QC, validation, discrepancy resolution, metadata harmonization, structured data organization, secure research data handling

Publications and Pre-prints

1. Aadrita Chatterjee, Brian Alvarez Alvarez, **Rakshya U Sharma**, Caroline Jonson, Heather M Wilkins, Judy Pa, Russell H Swerdlow, Kristine Yaffe, Alison M Goate, Shea J Andrews, Alzheimer's Disease Genetics Consortium, Alzheimer's Disease Neuroimaging Initiative. **"Evaluating the causal effect of mitochondrial dysfunction on Alzheimer's and Parkinson's disease using Polygenic Risk Scores and Mendelian Randomization"** *Alzheimer's and Dementia*, May 2026. DOI: <https://doi.org/10.1002/alz.71469>
2. Ben Weisburd, **Rakshya Sharma**, Villem Pata, Tiia Reimand, Vijay S. Ganesh, Christina Austin-Tse, Ikeoluwa Osei-Owusu, Emily O'Heir, Melanie O'Leary, Lynn Pais, Seth A. Stafki, Audrey L. Daugherty, Chiara Folland, Stojan Peric, Nagia Fahmy, Bjarne Udd, Magda Horáková, Anna Łusakowska, Rajanna Manoj, Atchayaram Nalini, Veronika Karcagi, Kiran Polavarapu, Hanns Lochmüller, Rita Horvath, Carsten G. Bönnemann, Sandra Donkervoort, Göknur Haliloğlu, Ozlem Herguner, Peter B. Kang, Gianina Ravenscroft, Nigel Laing, Hamish S. Scott, Ana Töpf, Volker Straub, Sander Pajusalu, Katrin Öunap, Grace Tiao, Heidi L. Rehm, Anne O'Donnell-Luria. **"Diagnosing missed cases of spinal muscular atrophy in genome, exome, and panel sequencing data sets"** *Genetics in Medicine*. April 2025. DOI: <https://doi.org/10.1016/j.gim.2024.101336>
3. Shea J Andrews, **Rakshya U Sharma**, Brendan A Mitchell, Tong Tong, Luke W. Bonham, Alan E Renton, Xiaoling Zhang, Marina Sirota, Duygu Tosun, Kristine Yaffe, the Alzheimer's Disease Neuroimaging Initiative. **"Integrative effects of Telomere Length, Epigenetic Age, and Mitochondrial DNA abundance in Alzheimer's Disease"** Pre-print *medRxiv*, September 2025. DOI: <https://doi.org/10.1101/2025.07.16.25331683>
4. Paulina Tolosa-Tort, Meri Okorie, Aadrita Chatterjee, Ana I Borieu, **Rakshya U. Sharma**, Shea J Andrews. **"Toward Personalized Medicine in AD/ADRD Through Genetic-Exposome Dementia Risk Assessments."** *npj Dementia*, September 2025

Manuscripts Under Preparation

1. **Rakshya U Sharma**, et. al., for ADGC and HABS-HD Study teams. **"Sex-Specific Genetic and Causal Effects of Modifiable Risk Factors on Alzheimer's Disease: Evidence from PRS and Mendelian Randomization."** , TBD

Conference Poster Presentations and Research Talks

1. **2025 MAC at Large meeting, UCSF**, "Sex differences in how genetic liability to modifiable traits influence Alzheimer's disease risk", Lightning talk.
2. **2025 PROPEL Scholars Symposium, UCSF**, "Polygenic Risk Scores for modifiable risk factors show sex-specific associations in Alzheimer's Disease", Research Talk.
3. **Rakshya U. Sharma**, Meri Okorie, Nia Spadoni, Aadrita Chatterjee, Kristine Yaffe, Shea J. Andrews for the Alzheimer's Disease Genetics Consortium. **"Polygenic Risk Scores for modifiable risk factors show sex-specific associations in Alzheimer's Disease"** 2025 *Alzheimer's Association International Conference, Toronto, Canada*, Poster

4. Aadrita Chatterjee, Brian Alvarez, **Rakshya U Sharma**, Heather Wilkins, Russell H Swerdlow, Kristine Yaffe, Shea J Andrews. **"Evaluating the Causal Effect of Mitochondrial Dysfunction on Alzheimer's and Parkinson's Disease using Polygenic Risk Scores and Mendelian Randomization"** *2025 Alzheimer's Association International Conference, Toronto, Canada, Poster*
5. Meri Okorie, Xiaqing Jiang, PhD, **Rakshya U. Sharma**, Caroline Jonson, PhD, Kristine Yaffe, PhD, Jennifer S. Yokoyama, PhD, Shea J. Andrews, PhD, for Healthy Aging Brain Study – Health Disparities. **"Orthogonal Contributions of Genetic and Clinical Risk Factors on Alzheimer's Disease and Neurodegenerative Pathogenesis"** *2025 Alzheimer's Association International Conference, Toronto, Canada, Poster*
6. **Rakshya U. Sharma**, Dennis Mulligan, Alexis Thornton, Angela N. Brooks. **"Building RNA Binding Protein Regulatory Signatures to Identify RNA Splicing Alterations in Cancer"** *2024 National Human Genome Research Institute (NHGRI) Annual Meeting, Seattle, WA, Poster*
7. **Rakshya U. Sharma**, Dennis Mulligan, Alexis Thornton, Angela N. Brooks. **"Building RNA Binding Protein Regulatory Signatures to Identify RNA Splicing Alterations in Cancer"** *2023 Advances in Genome Biology and Technology–Precision Health meeting, Annual Biomedical Research Conference for Minoritized Scientists, San Diego, CA, Poster*
8. **Rakshya U. Sharma**, Dennis Mulligan, Alexis Thornton, Angela N. Brooks. **"Building RNA Binding Protein Regulatory Signatures to Identify RNA Splicing Alterations in Cancer"** *2023 Annual Biomedical Research Conference for Minoritized Scientists (ABRCMS), Phoenix, AZ, Poster*
9. **Rakshya U. Sharma**, Dennis Mulligan, Alexis Thornton, Angela N. Brooks. **"Identifying the RNA Splicing Patterns in Cancer using Mutually Exclusive Splicing Analysis"** *2023 National Human Genome Research Institute Annual Meeting, Salt Lake City, UT, Poster*
10. **Rakshya U. Sharma**, Dennis Mulligan, Alexis Thornton, Angela N. Brooks. **"Identifying the RNA Splicing Patterns in Cancer using Mutually Exclusive Splicing Analysis"** *2022 SACNAS NDiSTEM Conference, San Juan, Puerto Rico, Poster*
11. **Rakshya U. Sharma**, Ben Weisburd, Heidi L. Rehm. **"Spinal Muscular Atrophy Diagnosis Using Whole Exome Sequencing Data"** *2022 Broad Summer Research Program Poster Symposium, Cambridge, MA, Poster*
12. **Rakshya U. Sharma**, Laura R. Nichols, Daniela Ushizima. **"Image Analysis of Biofuel Plant Root Composition in Drought Conditions"** *2021 Lawrence Berkeley National Lab Poster Symposium, Virtual, Poster*

Selected Software and Open-Source Tools

1. **Snakemake Executor Plugin for SGE/Wynton** Developed a custom executor plugin for scalable job submission and workflow execution on UCSF Wynton high-performance computing systems. **Tech:** Python, Snakemake, SGE, HPC **GitHub:** AndrewsLabUCSF/snakemake-executor-plugin-sge-wynton
2. **PRS-CS Auto Pipeline** Built a reproducible workflow for polygenic risk score generation, quality control, and downstream association analysis across large Alzheimer's disease cohorts. **Tech:** Python, R, Snakemake, Nextflow, Docker **GitHub:** AndrewsLabUCSF/PRS-CS-Auto

3. **RBPSig** Developed an RNA-seq analysis pipeline to generate RNA-binding protein regulatory signatures from ENCODE data and evaluate splicing-related alterations in cancer cohorts.
Tech: Python, STAR, MESA, RNA-seq **GitHub:** codewithrakshya/RBPSig

Relevant Strengths for Prevention and Population Health Research

Quantitative Research Support	Experience supporting collaborative research projects through data cleaning, harmonization, statistical analysis, reproducible reporting, manuscript-oriented summaries, and interpretation of complex biomedical datasets.
Clinical and Cohort Data Integration	Integrated genomic, clinical, and phenotypic metadata across 30+ datasets, supporting cohort quality control, association modeling, subgroup analysis, and reproducible downstream analyses.
Prevention-Relevant Methods	Experience with risk modeling, polygenic risk scores, Mendelian randomization, genetic correlation, sex-stratified analysis, and analysis of modifiable risk factors relevant to disease prevention and precision health.
Reproducible Research Infrastructure	Developed Snakemake and Nextflow workflows, containerized analysis environments, and HPC-compatible tools to improve reproducibility, scalability, and collaboration across research teams.
Health Equity and Global Health Interest	Interested in applying data science, epidemiology, and computational methods to prevention, aging, health disparities, global health, and implementation-focused research in diverse populations.

Research Interests

Computational biology; population health data science; epidemiologic data analysis; statistical genetics; prevention science; aging and neurodegenerative disease; health disparities; clinical and phenotypic data harmonization; EHR and cohort data integration; reproducible research workflows; implementation and evaluation support; multi-omics and high-dimensional data analysis; precision health; translational genomics; public health informatics; global health research.

Honors, Awards, and Research Programs

2024–Present	PROPEL Scholar , University of California, San Francisco
2022	Broad Summer Research Program , Broad Institute of MIT and Harvard
2021	Community College Internship , Lawrence Berkeley National Laboratory

Mentoring Experience

01/2025–04/2025	Research Mentor to Nia Spadoni, Community Health Advancing Next Generation Efforts San Francisco (CHANGE SF), University of California, San Francisco
08/2023–06/2024	Peer Mentor , Services for Transfer and Re-entry Students (STARS), University of California, Santa Cruz

Memberships

06/2024–Present	ISTAART , <i>International Society to Advance Alzheimer's Research and Treatment</i> .
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